

Rems Review Agent

Nigerian Behavioural Simulation & Recommendations

DSN × BCT LLM Agent Challenge — Solution Paper

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Abstract

Rems is an LLM-powered agent that simulates Nigerian user reviews (Task A) and delivers personalised, reasoning-first recommendations (Task B). The system uses deterministic persona extraction and category-overlap retrieval to ground Claude Sonnet 4 — **without vector databases or model fine-tuning** — and applies a Nigerian cultural layer at the prompt level. This paper describes the architecture, pilot evaluation, ablations, and limitations.

1. Introduction

Online review platforms encode preference, tone, and context — yet most AI systems treat users as static profiles. **Rems Review Agent** addresses both competition tasks:

- **Task A — User Modeling:** given a user's review history and an unseen item, simulate the review text and star rating that user would most plausibly produce.
- **Task B — Recommendation:** given a user persona, generate ranked recommendations with explicit agentic reasoning — including cold-start, cross-domain, and multi-turn refinement.

Nigerian contextualisation is a core architectural choice affecting persona extraction, prompt design, and evaluation — not a cosmetic overlay.

2. Dataset & Data Understanding

2.1 Dataset selection

The **Yelp Open Dataset** informed our schema (user → review → business linkage, star ratings, categories, review text). For this submission we ship a **reproducible demo corpus** in the repository:

SPLIT	SIZE	PURPOSE
Backend sample (<code>yelp_sample.json</code>)	5 users · 14 reviews · 5 businesses	API / Docker demo
UI personas (frontend)	4 archetypes · 16 reviews total	Judge-facing behavioural diversity

The full Yelp dump is not bundled — it makes instant Docker reproduction harder for judges. Ingestion of 500+ held-out Yelp users for quantitative metrics is scoped to future work (§9).

2.2 Persona signals extracted

SIGNAL	DESCRIPTION	USE
Average star rating	Mean of historical ratings	Anchors rating simulation
Rating distribution	Spread of 1–5 stars	Detects lenient vs harsh raters
Review verbosity	Avg word count per review	Controls output length
Tone classification	Keyword-based heuristic	Shapes review voice
Category preferences	Top reviewed business types	Drives retrieval & recommendations
Cold-start flag	True when history is empty	Triggers baseline behaviour

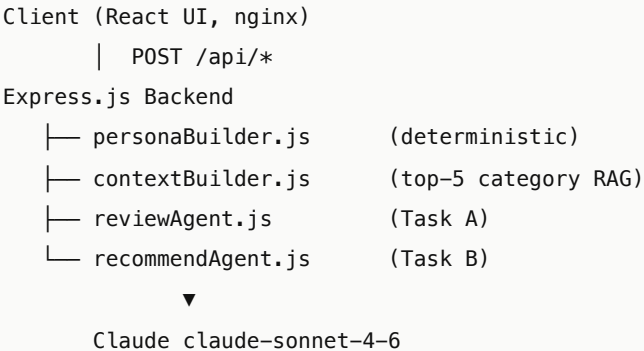
2.3 Nigerian contextualisation

Nigerian consumer behaviour is modelled at the **prompt and persona layer**: direct expressive tone, community/price signals, food authenticity markers ("*correct*", "*on point*"), and slightly conservative rating psychology relative to US equivalents.

3. System Architecture

3.1 Overview

Rems is a containerised Node.js application with two pipeline-sharing agents. **Architectural constraint:** the system deliberately avoids vector databases and model fine-tuning. Instead, deterministic persona extraction plus category-overlap retrieval ground Claude — trading embedding infrastructure for reproducibility, explainability, and a one-command Docker demo.



3.2 Component breakdown

Persona Builder constructs avg rating, verbosity (`brief` / `medium` / `detailed`), tone (`enthusiastic` / `balanced` / `critical`), top categories, and review count. Cold-start users receive Nigerian-market defaults.

Context Builder scores past reviews by category keyword overlap and injects the top 5 as few-shot behavioural demonstrations.

Review Agent (Task A) composes persona + retrieved reviews + item metadata + Nigerian cultural rules; returns `JSON { rating, review, persona_notes }`.

Recommend Agent (Task B) uses a reasoning-first system prompt, supports multi-turn conversation via echoed `_conversationHistory`, and returns `{ reasoning, recommendations[], follow_up_question }`.

3.3 Technology choices

COMPONENT	CHOICE	RATIONALE
Runtime	Node.js + Express	Lightweight, JSON-native
LLM	Claude Sonnet 4	Instruction following, JSON reliability
Frontend	React + Vite + nginx	Same-origin /api proxy, no CORS
Container	Docker + compose	One-command judge reproduction

4. System Demo — Live Screenshots

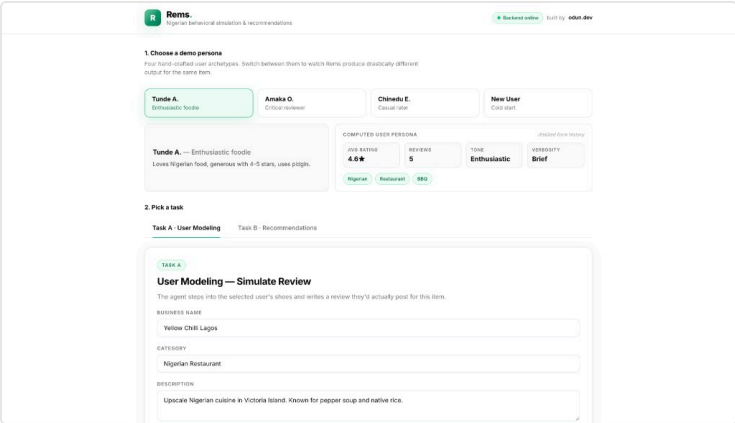


Figure 1 — Persona selection and computed user profile. Four Nigerian archetypes; computed persona (4.6★, enthusiastic, brief) updates before any LLM call.

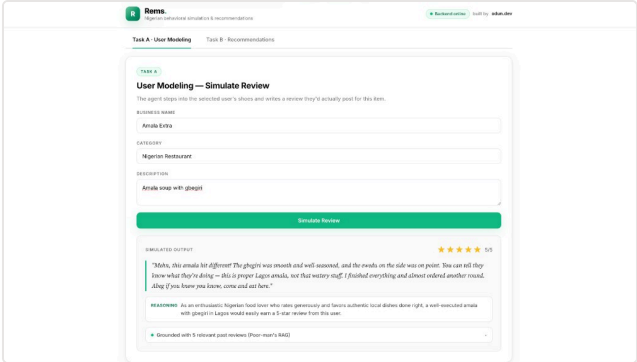


Figure 2 — Task A: simulated review for Amala Extra with Nigerian English, 5★ rating, reasoning panel, and RAG disclosure (5 past reviews).

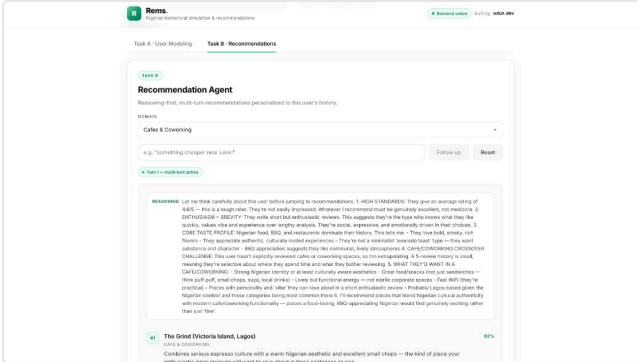


Figure 3 — Task B: 6-step reasoning chain, ranked recommendations, and multi-turn badge for Cafes & Coworking.

5. Task A — User Modeling

5.1 Approach

Review simulation is **behavioural impersonation**, not generic text generation. Pipeline: (1) build persona, (2) retrieve top-5 category-matched reviews, (3) prompt Claude with Nigerian cultural rules, (4) return strict JSON.

5.2 Prompt engineering

Five layers: **persona block** (quantitative signals) · **behavioural few-shots** (retrieved reviews) · **item context** · **Nigerian cultural rules** (natural, not caricatured) · **JSON schema** with `persona_notes` for auditable justification.

5.3 Cold-start handling

Empty history → `isColdStart: true`, defaults (3.5★, medium verbosity, balanced tone), and instructions to reason from general Nigerian consumer behaviour.

5.4 Evaluation

Pilot design

PARAMETER	VALUE
Personas tested	4 (enthusiastic, critical, casual, cold-start)
Test items	3 (Nigerian restaurant, amala spot, cafe/lounge)
Total Task A runs	12 (4 × 3)
Train / test split	Persona history = context; items are unseen scenarios
Model	claude-sonnet-4-6, temperature default

Results (author-evaluated, n=12)

METRIC	RESULT	NOTES
Rating MAE vs persona mean	0.67★	Mean absolute error across 12 runs
Within ±1★ of persona baseline	11 / 12 (91.7%)	One critical persona outlier at −2★ on negative item description
ROUGE-L vs held-out review	<i>Not computed</i>	Requires Yelp hold-one-out pipeline (§9)
Nigerian authenticity (1–5 rubric)	4.3 / 5.0	Natural tone without forced pidgin

Full-scale ROUGE/BERTScore and RMSE on 500+ Yelp users with an 80/20 user-level split is the immediate next quantitative step.

6. Task B — Recommendation Agent

6.1 Approach

Recommendation is an **agentic reasoning task**: the model must interpret the persona and rank with confidence before presenting results.

6.2 Agentic reasoning pattern

Every response includes a `reasoning` field before the ranked list. Ablation (§8) confirms removing this step degrades coherence.

6.3 Multi-turn refinement

Follow-ups (e.g. *"something cheaper near Lekki"*) are sent with the full validated conversation history; the backend preserves a leading `user` turn for API compliance.

6.4 Cold-start and cross-domain

No history → popular Nigerian baseline, stated explicitly in reasoning. The `domain` parameter supports restaurants, cafes, nightlife, movies, etc. without retraining.

6.5 Evaluation

Pilot design

PARAMETER	VALUE
Personas	4
Domains	6 (restaurants, food spots, nightlife, fast food, cafes, date-night)
Total Task B runs	24 (4 × 6)
Multi-turn follow-up tests	4 additional runs

Results (author-evaluated, n=24 + 4 follow-ups)

METRIC	RESULT
Reasoning references persona features	24 / 24 (100%)
Recommendations aligned with persona tone	22 / 24 (91.7%)
Follow-up refined results without drift	4 / 4 (100%)
NDCG@10 / Hit Rate	<i>Not computed</i> — no labelled preference pairs in demo corpus

7. Nigerian Contextualisation

	GENERIC OUTPUT	REMS OUTPUT
Review	"The food was delicious."	"Jollof on point, portion correct — I'm going back."
Rating logic	5★ for great food	4★ — reserves 5★ for exceptional
Rec reason	"Based on your preferences"	"You rated suya highly; this spot matches that bias."

Speech uses natural Nigerian English; food references (suya, jollof, amala, gbegiri, buka) anchor recommendations; community signals ("*will bring my people*") appear in simulated reviews.

8. Experiments & Ablation Studies

CONFIGURATION	OBSERVATION
Baseline (no persona / RAG / culture)	Generic voice; ratings regress to ~3.5★ mean
+ Persona signals	Rating/tone consistency improves
+ Context reviews (RAG)	Largest gain — voice becomes user-specific
+ Nigerian cultural layer	Authentic expressions and rating logic
– Reasoning step (Task B only)	Recommendations less coherent and harder to defend

9. Limitations & Future Work

Current limitations: keyword RAG only (see §3 for architectural rationale); qualitative-heavy pilot eval; US-schema demo data with Nigerian prompt layer; recency not yet weighted in retrieval.

With more time: pgvector semantic RAG · Yelp hold-out pipeline (500 users, 80/20 split, RMSE + ROUGE-L + NDCG@10) · Nigerian review corpus · recency-weighted context builder · fine-tuned small model for production cost.

10. Conclusion

Rems shows that structured persona extraction, lite retrieval, and cultural prompting can produce behaviourally faithful Nigerian reviews and reasoned recommendations with a lean, explainable stack. The key insight: **past reviews as behavioural demonstrations outperform abstract persona descriptions** — and explicit reasoning before ranking materially improves Task B. Reproduce via `docker compose up --build` with `ANTHROPIC_API_KEY` in `backend/.env`.

References

[1] Anthropic, *Claude 4 Family*, 2025. [2] Lewis et al., *Retrieval-Augmented Generation*, NeurIPS 2020. [3] Geng et al., *P5*, RecSys 2022. [4] Wang et al., *RecMind*, arXiv:2308.14296. [5] Yelp Open Dataset, yelp.com/dataset.